**Project Development Phase**
**Model Performance Test**

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| Date | 31 January 2026 |
| Team ID | LTVIP2026TMIDS49942 |
| Project Name | Exploratory analysis of rain fall data in India for agriculture |
| Maximum Marks | 10 Marks |

**Model Performance Testing:**

Project team shall fill the following information in model performance testing template.

| S.No. | Parameter | Values | Screenshot |
|---|---|---|---|
| 1 | **Metrics** | **Regression Model:**<br>MAE = **0.32**<br>MSE = **0.21**<br>RMSE = **0.46**<br>$R^2$ Score = **0.81**<br>**Classification Model:**<br>Confusion Matrix =<br>[[21220 857 61]<br>[3598 2639 75]<br>[358 74 310]<br>Accuracy Score = **0.83**<br>Classification Report = Precision = **0.81**<br>Recall = **0.83**<br>F1-Score = **0.81** | [[21220 857 61]<br>[ 3598 2639 75]<br>[ 258 74 310]]<br>Accuracy: 0.8307782208167194<br>Precesion: 0.8196736660656065<br>Recall: 0.8307782208167194<br>F1-score: 0.8124789140764411 |
| 2 | **Tune the Model** | **Hyperparameter Tuning:** Grid Search was used to optimize model parameters such as number of trees and maximum depth. **Validation Method:** KFold Cross Validation (k = 5) | |